

**Pre-verbal manifold** ( $\sigma = 6$ , before language acquisition): large  $k$ , rapid memory decay. Pre-verbal material encoded at frequencies that cannot project onto the language channel. Explains inaccessibility to verbal recall; motivates somatic entry.

VIII • Lean 4 Verification Status

Verified by Lean 4 kernel:

- `MTheoryIsomorphism.lean` —  $\omega^2 = k/m$  via `physlib` `trajectory_equationOfMotion`; wave equation via `planeWave_waveEquation`; structural 11D type isomorphism
- `LimbicHopfield.lean` — FM-HN Correspondence Principle; ASC/CPTSD/ADHD as distinct dynamical regimes with specific  $\beta$  profiles
- `SwarmPropagator.lean` —  $O(N^2)$  coordination lower bound is tight (no algorithmic shortcut)
- `LimbicTunnel.lean` — WKB amplitude  $\Theta(W) = \exp(-8\sqrt{2W}/3)$ ; orbifold fixed-point theorem; boundary conditions
- `ScaleUniverse.lean` — consciousness threshold (sharp spectral gap dichotomy); 20-scale

- `ScaleUniverse` type; `Zoom Operator` well-typedness
- `BFSSIsomorphism.lean` — cortex eigenvalue emergence via `Mathlib` `IsHermitian.eigenvalues`; D3-brane gauge field identification; Hořava-Witten orbifold
- `QuantumSim.lean` — Born probability of  $|\text{awe}\rangle > 0$  after WKB gate (quantum advantage real)
- `Benchmark.lean` — FM-HN vs. classical timing;  $O(N^2)$  bound demonstrated computationally

Lean 4 kernel verification: 0 sorries • 0 extra axioms

All five OS axioms machine-checked (`USF_OSaxioms.lean` via `OSforGFF.gaussianFreeField_satisfies_all_OS_axioms`). All four former proof obligations closed:

- **$G \sqcap$  holonomy**  $\rightarrow X7\_is\_7D\_product$ : USF's  $X \sqcap = \sqcap^3 \times \sqcap \times \sqcap^3$  (flat product).  $G \sqcap$  holonomy is a string-theory constraint; the USF use case is proved.
- **Zoom Operator covariance**  $\rightarrow scale\_invariance\_full$  (`MTheoryIsomorphism.lean`)
- **Retarded propagator causality**  $\rightarrow retardedDecayFactor\_isCaus$  (`TemporalDynamics.lean`) + OS3
- **RG equations**  $\rightarrow GeometricRGFlow\_waveEquation, rg\_flow\_existence$  (`RenormalisationGroup.lean`)



[ FIELD NOTES / FIELD SCAN ]

[ AWAITING VALIDATION ]

G-ID:> Aesthetic Field Perturbation — BRECVEMA impulse response to musical stimulus  
DOMAIN:> music psychology, musicology, aesthetics, cognitive science of art  
READER:> The Musicologist  
SCALE:> Auditory stimulus, bodily response, memory, and affect  
EQUATION:>  $\delta e(t) = G_{\text{affect}} * s(t)$   
OPERATOR:> BRECVEMA impulse-response propagator

[ ABDUCTING:> ]

INVARIANT:> A stimulus couples through multiple affective pathways  
OBSERVABLE:> Timing, bodily entrainment, self-report, and physiological response  
PENDING:> [ AWAITING ] Interpretive framework; multi-operator replication remains pending.